

TOPIC NAME: Summer 25 (A)

DAY: _____

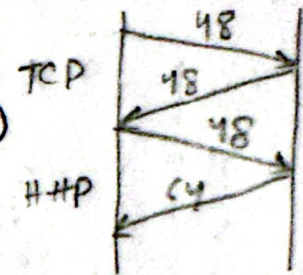
Azmari Sultana

TIME: _____

DATE: / /

(c)

$$\begin{aligned} \text{i) Total RTT} &= (48 + 48) + 10(48 + 64) \\ &= 1104 \text{ ms} \end{aligned}$$



$$\begin{aligned} \text{ii) Transmission time} &= 123 + 3 \times 520 + 5 \times 333 \\ &= 3348 \text{ ms} \end{aligned}$$

$$\text{Total time required} = 1104 + 3348 = 4452 \text{ ms}$$

$$\text{iii) Total time} = 1104 + \frac{3348}{2} = 2778 \text{ ms}$$

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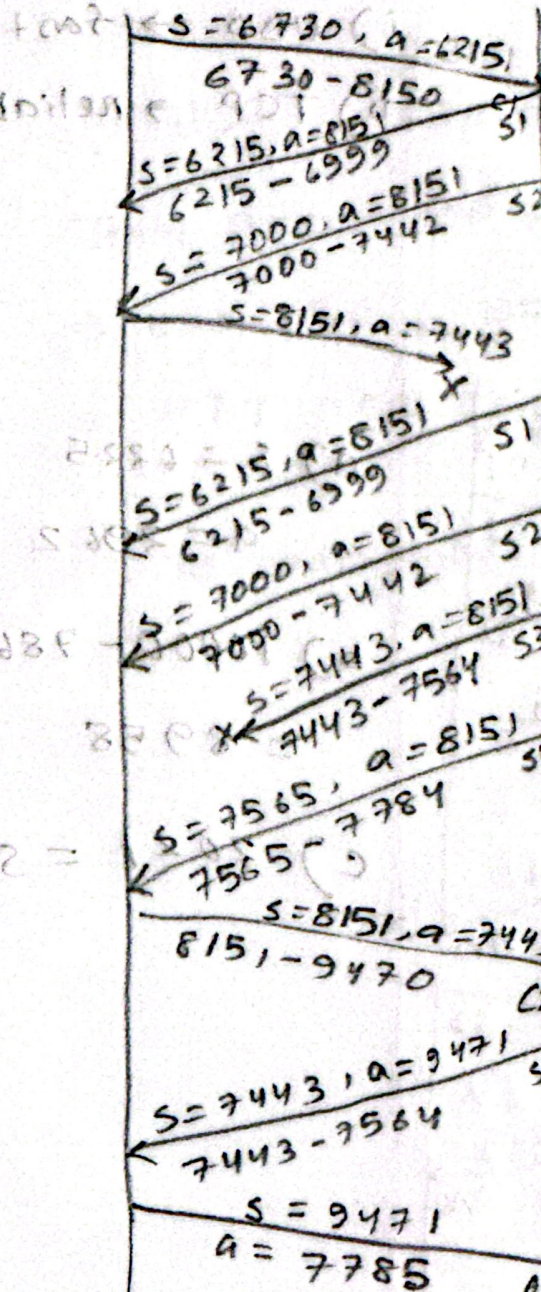
DAY: _____

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⑦ client

server



a) 7000 - 6215

= 785

b) S=8151

(A) a=7443

c) 3200 - 785

- 443 - 122 - 220

= 1630

GOOD LUCK

TOPIC NAME: Spring 25 (A)

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⑦ DNS look up takes 1 RTT = 50 ms

$$\text{Total object} = 23 + 1 = 24$$

$$a) \text{ Total RTT} = \text{DNS RTT} + (\text{number of object} + 1) \times (\text{RTT per object})$$

$$\Rightarrow 850 = 50 + (24 + 1) \times \text{RTT}$$

$$\therefore \text{RTT per object} = 32 \text{ ms}$$

$$b) \text{ File transmission time} = \frac{(5 \times 6) \times 8 + (19 \times 2) \times 8}{200}$$

$$= 2.72 \text{ sec}$$

$$= 2720 \text{ ms}$$

$$\text{TT} + \text{TTR} = \text{total time}$$

$$850 + 2720 =$$

$$3570 \text{ ms}$$

$$\frac{\text{size of object}}{\text{bandwidth}} = \text{transmission time}$$

$$(8 \times 6) \times \frac{1}{2} + 19 \times 1 = \text{total time}$$

$$2720 \text{ ms}$$

Flag

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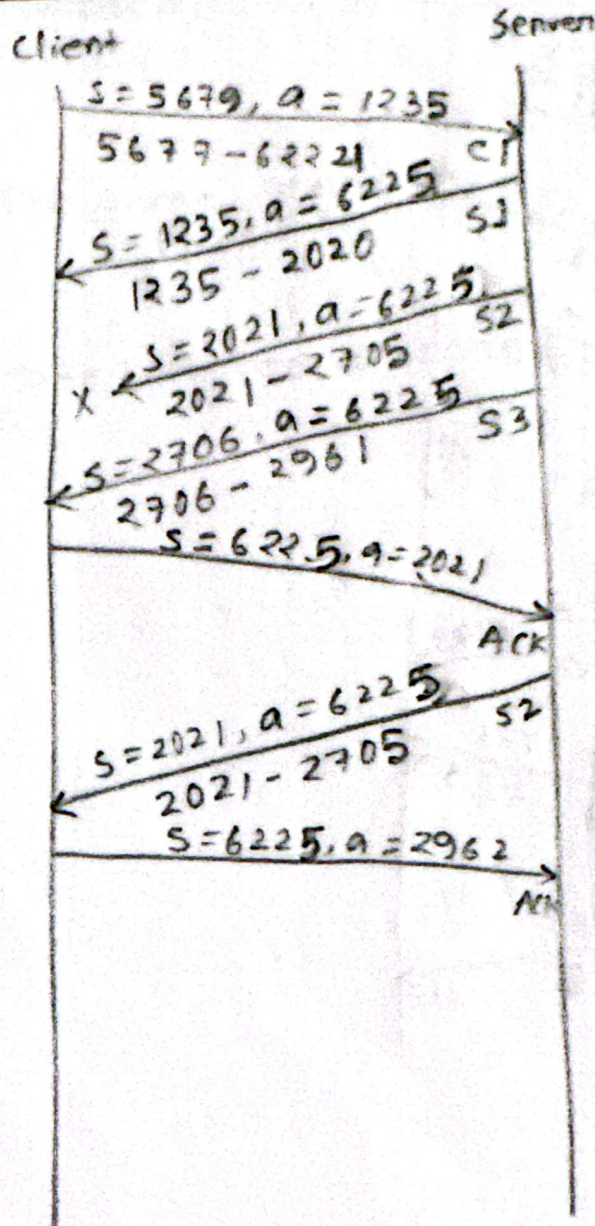
HTTP Method type:

Get, post, head,
put/patch
delete

Transport layer protocol

- i) UDP → fast
- ii) TCP → reliable

Spring 25 (A)



a) $S = 6225$

$A = 2962$

b) $10000 - 786 - 256$
 $= 8958$

c) $2021 = S_5$

GOOD LUCK

TOPIC NAME: FALL 24 (A)

DAY: _____

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- ⑥ non-persistent, obj-size = 4 MB, $TCP RTT = 35 ms$
Object = 34 (excluding base)
Object = 34 + 1 = 35, HTTP RTT = 30 ms
Speed 64 Mbps

a) Total RTT = TCP RTT X Object + HTTP RTT X Object
 $= 35 \times 35 + 30 \times 35$

+ Total RTT = 2275 ms

b) Total File transmission time = $\frac{35 \times 4 \times 8}{64}$

= 17.5 s

= 17500 ms

Object = 35

Object = 35

Total RTT = 17.5 s X 35 = 612.5 s

GOOD LUCK

TOPIC NAME: _____

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⑦ i)

$$\begin{aligned}\text{average response time} &= 0.4 \times 15 + 0.3 \times 30 \\ &\quad + 0.3 (100 + 200) \\ &= 105 \text{ ms}\end{aligned}$$

TOPIC NAME: Fall 2024 (A)

DAY: _____

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⑧

Client

Server

$S = 1911$

$S = 1533, a = 1912$

$S = 1912, a = 1534$

$1912 - 2332$

$S = 1534, a = 2333$

$S = 1534, a = 1793$

$S = 1794, a = 2333$

$S = 1794, a = 2013$

$S = 2333, a = 2014$

$2333 - 2652$

$S = 1534, a = 2014$

$1534 - 1793$

$S = 2653, a = 2014$

$2653 - 2763$

$S = 2014, a = 2764$

ack

ii) $S = 2333$

$a = 2014$

iii) $12000 - 421 - 111$

$= 11468$

GOOD LUCK

TOPIC NAME: Summer 24 (A) DAY:

TIME: DATE: / /

⑦ non-persistent, obj-size = 1.5 Mb, delay = 7 ms

$$RTT = 34 \times 2 = 68 \text{ ms}^{\text{34}}, \text{ Speed} = 60 \text{ Mbps}$$

$$RTT = 0.034 \text{ s}$$

i) Transmission time per object = size / bandwidth

$$= \frac{1.5 \text{ Mb}}{60 \text{ Mbps}} = 0.025 \text{ s} = 25 \text{ ms}$$

Per object time excluding RTT = waiting delay + transmission time

$$= 7 + 25 = 32 \text{ ms}$$

$$= 0.032 \text{ s}$$

$$\text{Total (excluding RTT)} = x \cdot 0.032 = 1.184 \text{ sec}$$

$$\therefore x = \frac{1.184}{0.032} = 37 \text{ objects}$$

ii) RTT per small packet = 34 ms = 0.034 sec

Non-persistent $\rightarrow$ 2 RTTs per object

$$\text{Object} = 37$$

$$\text{Total RTT} = 37 \times 2 \times 0.034 = 2.516 \text{ s}$$

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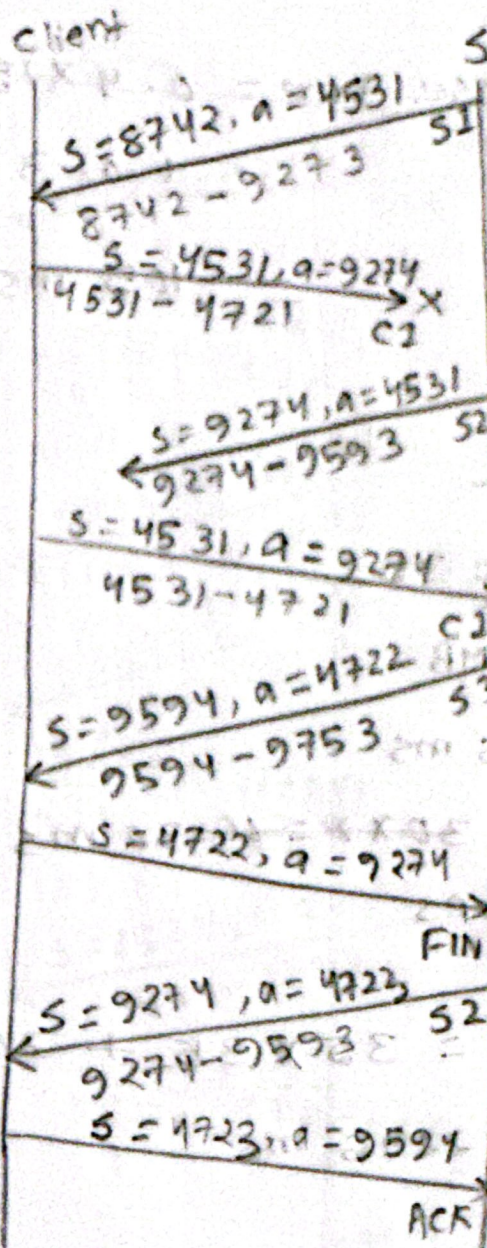
⑧

$C1 = 191$

$S1 = 532$

$S2 = 320$

$S3 = 160$



i) $S = 4722, a = 9274$

ii) $S = 4723, a = 9594$

iii) $14000 - 191 = 13809$

GOOD LUCK

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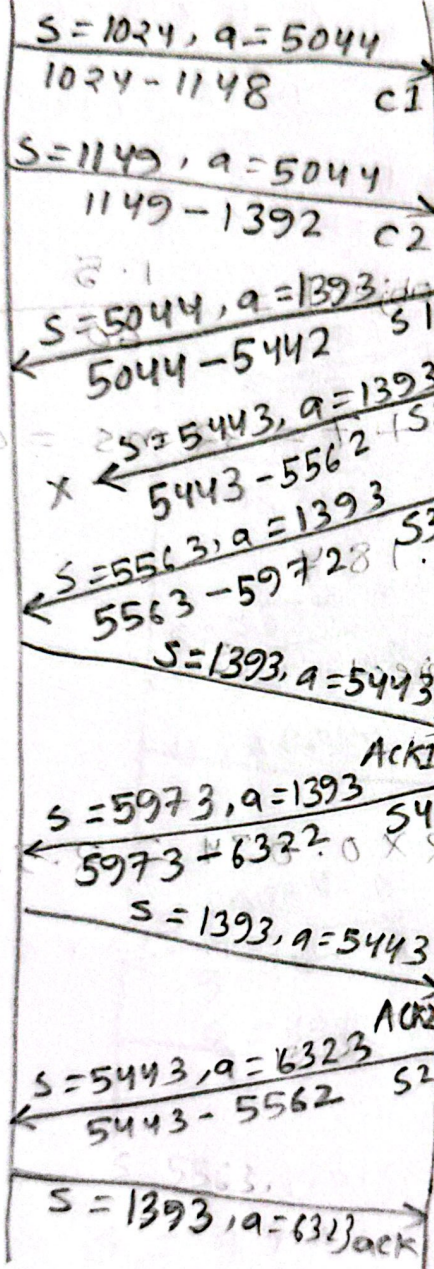
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⑧

client

Server



first 212 1799 non

c1 = 125

c2 = 244

s1 = 399

s2 = 120

s3 = 410

s4 = 350

i) $s = 5563$

ii) $a = 5443$

iii) $a = 6323$

FE = X

FE = X

FE = TTA Intat (ii)

GOOD LUCK

TOPIC NAME: Summary

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⑦ a)

one-way time = 15 ms

$$RTT = 15 + 15 = 30 \text{ ms}$$

$$\text{Total RTT} = 480 \text{ ms}$$

For persistent:

$$\text{Total RTT} = (1 + \text{object}) RTT$$

$$\Rightarrow 480 = (1 + \text{obj}) 30$$

$$\Rightarrow \text{obj} = 15$$

b) Total file transmission time = $\frac{15 \times 10 \times 8}{80}$

$$= 15 \text{ ms}$$

GOOD LUCK